

Reverse Engineering Report

Heat Pump Tumble Dryer Part

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Virtual Product Design

Introduction

How 3D Scanning Works

3D scanning is a **non-contact measurement process** used to digitally capture the shape and geometry of real-world objects. It operates by projecting light (such as **laser lines** or **structured light patterns**) onto the object's surface and detecting its deformation through one or more **cameras**.

This data is processed to form a **point cloud** or **polygon mesh**, representing the object's 3D geometry. In triangulation-based systems, the angle between the **laser source**, **object**, and **camera** allows for highly accurate depth calculation. Modern scanners, such as handheld laser scanners, use **self-positioning** or **optical tracking**, eliminating the need for fixed reference points.

The scan data is typically exported as an **STL file**, which can be further processed in CAD software for reverse engineering or 3D printing. Applications include **quality inspection**, **rapid prototyping**, and **digital archiving**, making 3D scanning an essential tool in **virtual product development**.



Figure 1: 3D scanning process

Task Description

This group project aimed at the reverse engineering of a part of a heat pump tumble dryer. Namely, a 0107- fascia control panel.

The important objectives were:

1. To scan physical components through 3D scanner.
2. To re-make the injection-molded parts in SolidWorks, relying on the mesh.
3. Write report about the experience

This whole project demonstrates how there are other techniques that may be used to redesign any component so that it may be remolded even without having the actual physical object.

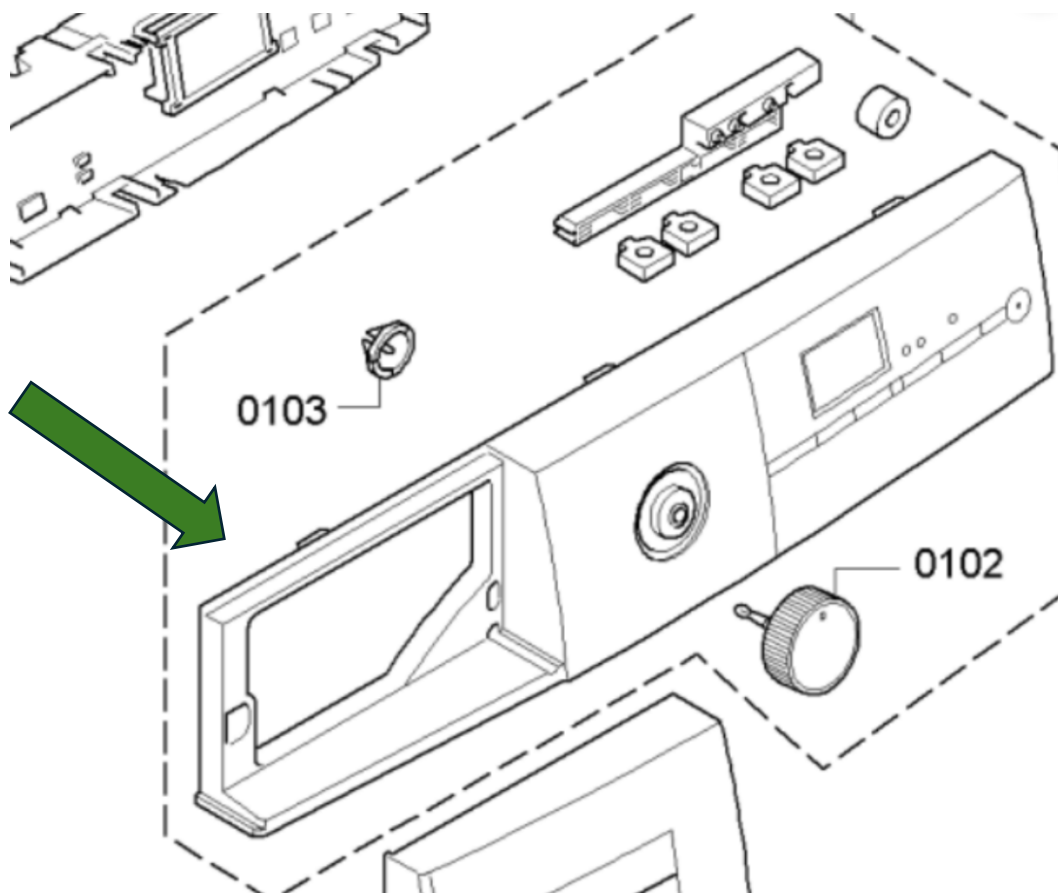


Figure 2: Control Panel

General Settings

Hardware:

3D Scanner: HandySCAN 300 by CREAFORM

- Measurement rate: 205,000 measurements/s
- Scanning area: 225 × 250 mm
- Light source: 3 red laser crosses
- Laser class: 2M (eye-safe)
- Resolution: 0.1 mm
- Accuracy: up to 0.04 mm
- Volumetric accuracy: 0.02 mm + 0.1 mm/m
- Stand-off distance: 300 mm
- Depth of field: 250 mm
- Recommended part size: 0.1 – 4 m

Software:

- VXelements (CREAFORM) – for scan process and mesh generation
- SolidWorks 2024 – for CAD modeling, sheet metal, and mold design. ScanTo3D was also used as an add on to import the mesh as graphics body.



Figure 3: HandySCAN 300

Approach

We began the reverse engineering process by selecting the 0107 panel. We then began taking turns to scan the panel using the HandySCAN 300 handheld 3D scanner. The scanner captured detailed geometric data, which was converted to mesh file and was then exported in STL format.

The STL mesh file was imported into SolidWorks as a **Graphics Body**. After it was imported, I then used ScanTo3D add in to begin mesh preparation. After mesh prep, I used surface wizard to try and extract some surfaces. This was very unpleasant since it felt as like the mesh was losing resolution with each step. Additionally, due to the complexity and size, SolidWorks crashed a couple of times, so I had to reduce the number of faces of the mesh and save after every change I made.

After that, I tried to create an origin by looking for a straight surface or use a point so I can make planes. I also used ScanTo3D's mesh prep to make the planes and origin using the mesh data. Then, I tried to reconstruct the surface geometry of the whole model as best as I could. This step was particularly difficult due to the geometry of the panel. Its surfaces were curved and some of the bosses had lost a lot of quality.

I remade some of these as simple as I can using the scanned mesh data. The resolution was not as precise as expected. This also showed a key limitation to this process, our scan shows our scan may not have been detailed enough, highlighting one of the potential downsides of 3D scanning technology.

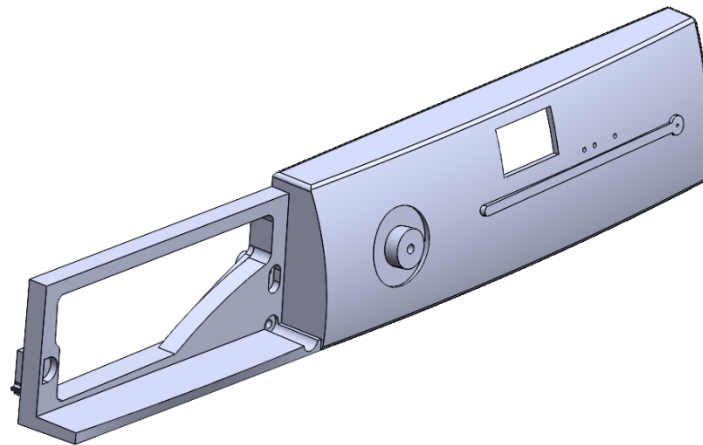


Figure 4: CAD model of the panel

Mold Creation

For the mold creation, I used the Cavity Tool in SolidWorks assembly. I first imported the model into assembly, then after I did, I inserted a new component but sketched on the part to create a cavity. After I sketched a rectangle, I used the Cavity tool built in SolidWorks to make the cavity. After selecting the settings, my cavity was made, I then also made a core in a similar manor.

Conclusion

We were able to reconstruct a simplified but also a representative version of the panel using the scanned surfaces of the geometry. This experience emphasized the importance of scan quality and resolution in reverse engineering.

References

- [Unit2: 3D Scanning](#)
- [Virtual Product Design Task](#)